

SHORT-TEST 2026-27
CLASS XI – ARTIFICIAL INTELLIGENCE

Maximum Marks: 35

Time Allowed: One and Half Hour

Instructions:

- Attempt all questions.
- Marks are indicated in brackets next to each question.

Section A

Question 1: Write the correct answer only. example: [X.(c) Answer] [1 × 5 = 5 marks]

- I) Which of the following is an application of Artificial Intelligence?
(a) Calculator (b) Weather Prediction (c) Ceiling Fan (d) Flash Drive
- II) Which type of problem always gives the same output for the same input?
(a) Probabilistic (b) Deterministic (c) Random (d) Intelligent
- III) Which operator is used to check equality in Python?
(a) = (b) == (c) != (d) i=
- IV) Which logic gate produces HIGH output only when all inputs are HIGH?
(a) OR (b) NAND (c) XOR (d) AND
- V) Which of the following matrix operations is possible only when the number of columns of the first matrix equals the number of rows of the second matrix?
(a) Addition (b) Subtraction (c) Multiplication (d) Transpose

Section B

Question 2: Answer the following in brief.

[1 × 10 = 10 marks]

- a) Define Artificial Intelligence.
- b) State any TWO applications of AI in healthcare.
- c) Differentiate between deterministic and probabilistic problems with one example each.
- d) What is the purpose of Data Acquisition in the AI Project Framework?
- e) Write the output of the following Python code.

```
1 x = 8
2
3 if x > 5:
4     if x < 10:
5         print("Kobra!")
6     else:
7         print("Anaconda!")
8 else:
9     print("Python!")
```

- f) Name any TWO Python data types.
- g) Identify the type of the following matrices.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 3 & 5 \end{bmatrix}$$

- h) Find the transpose of

$$\begin{bmatrix} 2 & 5 \\ 1 & 4 \end{bmatrix}$$

- i) Write the truth table of the OR gate.
- j) What is Cybersecurity? Why is it important in AI?

Section C

(Long Answer Type Questions)

[4 × 5 = 20 marks]

Question 3

Write a python program to determine whether a given year is a leap year or not.

[4]

Question 4

Explain the five stages of the AI Project Framework with suitable examples. Draw a flow chart of a successful AI project.

[4]

Question 5

- a) Compare Human Intelligence and Artificial Intelligence in any FOUR points.
- b) Explain any TWO real-life applications of Artificial Intelligence.

[2]

[2]

Question 6

- a) Draw the truth table and the expression of XOR gate.
- b) Explain the difference between AND gate and OR gate with Boolean expressions.

[2]

[2]

Question 7

- a) If

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

Show that the addition of matrices is commutative .

[2]

- b) Write a Python program that accepts a number from the user and checks whether it is **Even** or **Odd** using an **if-else** statement.

[2]